

Qwen3.8-27B across the effort knob

Alibaba’s open-weights 27B on the full v3 suite: twenty-three real merged post-cutoff PRs across five languages, at low, medium, and xhigh reasoning effort, through the first-party DashScope API. The smaller sibling of the flagship Qwen3.8-Max measured in Report No. 12.

ABSTRACT

Turning up the reasoning dial does not help this model, and eventually hurts it. Qwen3.8-27B scores 82.6% pass@1 at both low and medium effort, then falls to 73.9% at xhigh: flat, then backward. The extra effort is not free time, either. xhigh runs a median 17 minutes per task against 7 at low, 2.4 times slower for a worse result, and every one of its failures is an unfinished run that reasoned past the time budget rather than a wrong answer, including two tasks lower effort had solved in minutes. The shape echoes the flagship: Report No. 12 found Qwen3.8-Max running the same knob backward, 81.2% to 55.1%, a 26-point collapse. The open 27B is far more robust, dropping 8.7 points rather than 26, and it matches the flagship at low effort. On this suite the reasoning knob is at best inert and at worst a liability for the Qwen 3.8 family, and the small open model wears it better than the large hosted one.

TABLE 1. pass@1 by effort level (mean per-task success rate; single pass; ±1 stderr)

Effort	pass@1	Solved	Median time/task	Cost/task
low	82.6% ±8.1	19/23	7.1 min	\$0.71
medium	82.6% ±8.1	19/23	6.8 min	\$0.48
xhigh	73.9% ±9.4	17/23	17.1 min	\$0.69

One attempt per task per level (the reportable comparison). The low column additionally has three attempts per task, whose mean is 76.8% ±8.2, so its single-pass 82.6% is a slightly lucky draw and the true low-vs-medium gap is smaller still. Cost is per-task at DashScope list price (\$0.50/\$3.00 per M input/output, implicit cache \$0.10); the total sweep was \$82.71 over 129 runs.

TABLE 2. The reasoning knob across the Qwen 3.8 family

Model	low	medium	xhigh	low to xhigh
Qwen3.8-27B (open)	82.6%	82.6%	73.9%	−8.7
Qwen3.8-Max (Report No. 12)	81.2%	71.0%	55.1%	−26.1

Both models run the knob backward; xhigh is the shipped default for the family. Report No. 12 measured the Max at three attempts per task; the direction, not the exact single-pass value, is the comparison.

FINDINGS

- Effort is inert, then it runs out the clock.** Low and medium tie at 82.6%; xhigh drops to 73.9%. The extra reasoning improved nothing, and the drop is not wrong answers: all six xhigh failures were unfinished runs cut off at a step or wall-clock budget, none finished-but-wrong. xhigh did not make the model less accurate, only too slow to ship a patch in a fixed budget, the same failures-change-kind pattern Reports No. 12 and 14 found.
- xhigh costs time for a worse result.** A median 17.1 minutes per task at xhigh against 7.1 at low, 2.4 times slower. One task, *pennylane-trotter-fragmented*, ran the full 60-minute budget at xhigh and still failed, having been solvable at lower effort.
- Higher effort ran two quick wins into the ground.** *flask-teardown-robust* and *semver-inc-dotted-prerelease* were solved in minutes at low and medium. At xhigh the model reasoned past 20 minutes and 70 to 120 steps and never shipped a finished patch on either.
- The open model is more robust than the flagship.** Qwen3.8-Max lost 26 points across the same knob (Report No. 12); the 27B loses 8.7. At low effort the open 27B (82.6%) matches the hosted Max (81.2%) on this suite.
- Four tasks resist every setting.** *aiohttp-upgrade-deferred*, *networkx-leiden-communities*, *pennylane-trotter-fragmented*, and *sqlglot-canonicalize-internal-names* go unsolved at low, medium, and xhigh alike.

CAVEATS

This is a single-pass measurement, so the per-column standard errors are wide (±8 to 9 points) and low versus medium is a statistical tie; the reliable signals are the xhigh drop, the 2.4x slower runtime, and the two regressed tasks, which point the same direction. DashScope does not publish the serving quantization for the open 27B, so "the model" here means whatever Alibaba serves first-party, comparable to how Report No. 12 treated the Max. Costs are unusually high for a small model because the 1M-token context window lets the agent re-send its full growing transcript every step with no truncation, and implicit caching did not engage; a local run of the same weights, capped at a 32K window, used a fraction of the tokens. The comparison to Report No. 12 spans a single-pass column against a three-pass one, so only the direction transfers, not the exact gap.

METHOD

VulcanBench v3: 23 tasks from real merged post-cutoff PRs (Python 9, TypeScript 4, Rust 4, Go 3, JavaScript 3), graded by deterministic hidden tests in Docker; model judges disabled. Runs executed 2026-08-21 against DashScope’s international OpenAI-compatible endpoint (qwen3.8-27b), with `reasoning_effort` set to low, medium, and xhigh, the documented enum for the model (there is no "high"). Each level is one attempt per task; the low level was run three times and reports its first attempt here for symmetry, with the three-pass mean noted. pass@1 is the mean per-task success rate. A per-request ceiling of 1200 seconds was applied so a single xhigh reasoning turn on a hard task is not cut off mid-thought; the run’s overall wall-clock budget still bounds each task. This is the open-weights counterpart to Report No. 12’s Qwen3.8-Max on the same suite and the same reasoning dial.